

User Manual

COS 301 - Capstone 2026

Capstone Project:

EquityLens By The Big Five



Project Name: EquityLens

Client: Amazon Web Services (AWS)

Team: The Big Five (TB5)

Team members:

- Abdulrahman Sabah (Team Lead) - u24566170
- Joshua Heath - u23541475
- Antony van Straten - u24590739
- Jonty Honey - u23536862

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Welcome to EquityLens

EquityLens is a free web application that helps South African retail investors understand what they own. You upload your investment portfolio and EquityLens turns it into a clear picture of what your investments are worth, how healthy your portfolio is, how you are performing against the JSE, which news stories moved your stocks, and what deserves your attention. An AI assistant is built in so you can ask questions about your own portfolio in plain language and get plain-language answers.

- EquityLens never gives financial advice. It shows you information and explains what the numbers mean, but it will not tell you what to buy or sell.
- EquityLens never asks for your brokerage login. Your portfolio enters the system only through files you upload or numbers you type. The system cannot touch your actual investments and only analyses a copy of the information you give it.

Before You Start

You need three things:

- A modern web browser - Chrome, Firefox, Edge or Safari, on a computer, tablet or phone. The interface adapts to all screen sizes.
- An email address you can access, for the verification code that activates your account.
- An authenticator app on your phone, such as Google Authenticator or Microsoft Authenticator, for the two-step verification that protects your account.

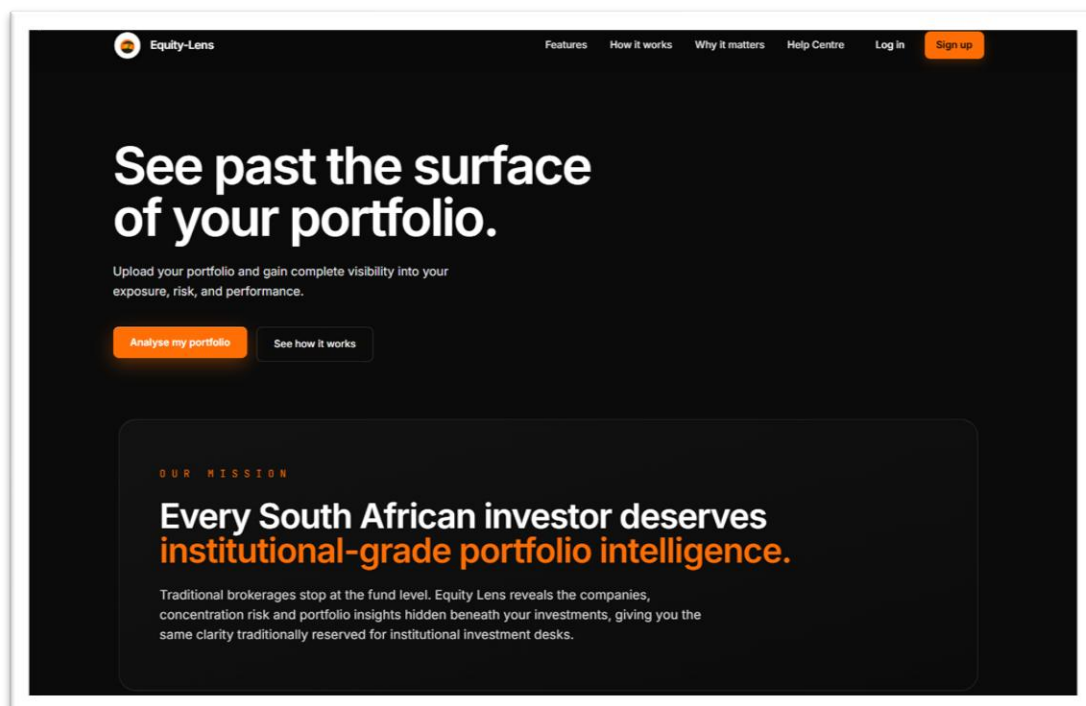
The Landing Page

Your first sight of EquityLens is its landing page headlined “See past the surface of your portfolio.” It exists to show you what the product does and why it matters before you have created an account.

From the navigation bar at the top, you can jump to:

- How it works - an interactive portfolio simulator. Flip between three example portfolios and see how Equity Lens’s analysis exposes what each one really holds underneath its funds including how supposedly different funds can quietly hold the same few companies.
- Features - the four things EquityLens does for you, shows the underlying companies in your funds, scores your portfolio’s health, ties news to your holdings, and lets you ask the AI assistant directly.
- Why it matters - a walk-through concentration risk with illustrative figures showing how a portfolio that looks diversified can fall hard when its overlapping holdings fall together.
- Help Centre - guides and frequently asked questions.
- Sign in/Get started - for returning and new users respectively.

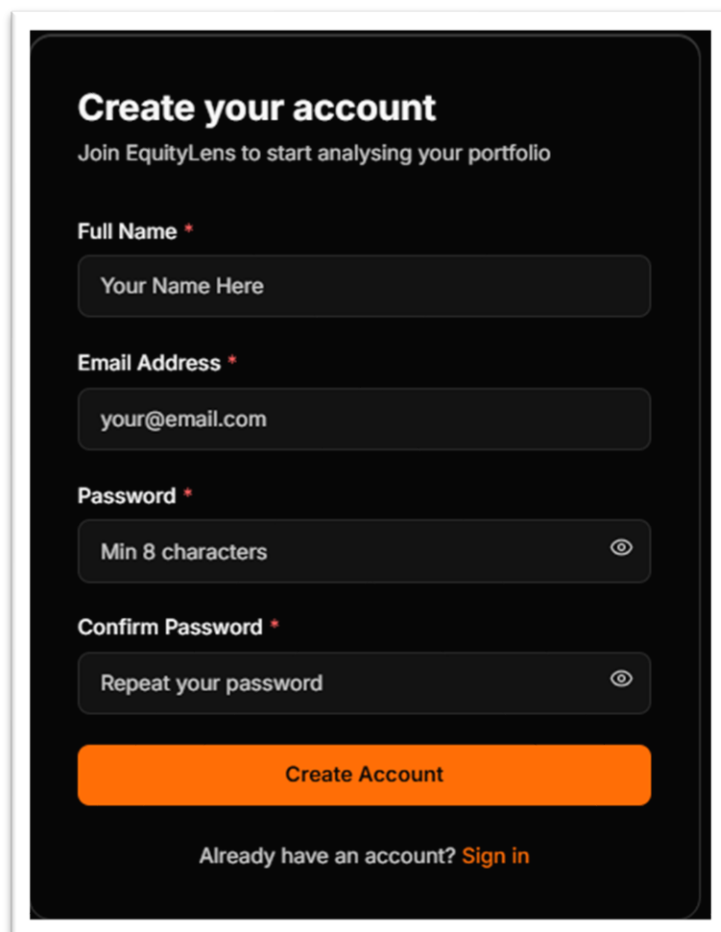
Select Get started (or analyse my portfolio button) to create your account.



Creating Your Account

1. From the landing page select Get started. The registration page opens.
2. Enter your full name, email address, and a password, then retype the password again in confirm password. Your password must be at least 8 characters and include an uppercase letter, a lowercase letter, a number and a special character.
3. Select Register. EquityLens emails a 6-digit verification code to the address you entered.
4. On the verification page, enter the 6-digit code from the email and submit it.
5. You'll see a confirmation that your account is active, and you'll be taken to the sign-in page.

If the code doesn't arrive within a couple of minutes, check your spam or junk folder. If the code is rejected as invalid or expired, the field clears and you can try again.



The image shows a dark-themed registration form titled "Create your account". Below the title is a subtitle: "Join EquityLens to start analysing your portfolio". The form contains four input fields, each with a label and a red asterisk indicating a required field:

- Full Name ***: The input field contains the placeholder text "Your Name Here".
- Email Address ***: The input field contains the placeholder text "your@email.com".
- Password ***: The input field contains the placeholder text "Min 8 characters" and has an eye icon on the right to toggle password visibility.
- Confirm Password ***: The input field contains the placeholder text "Repeat your password" and has an eye icon on the right to toggle password visibility.

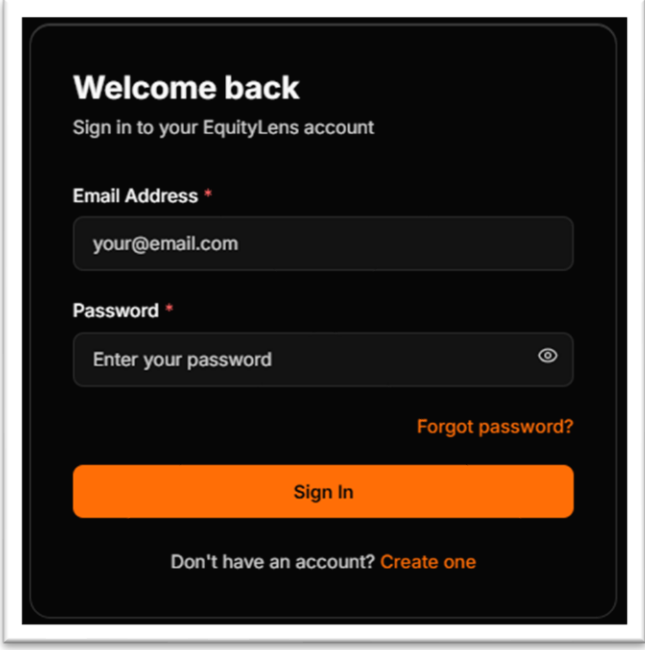
Below the input fields is a large orange button labeled "Create Account". At the bottom of the form, there is a link that says "Already have an account? Sign in", where "Sign in" is in orange text.

Signing In with Two-Step Verification

EquityLens protects your account with two steps: your password, plus a 6-digit code from an authenticator app on your phone. A stolen password alone is not enough to reach your data.

The first time you sign in:

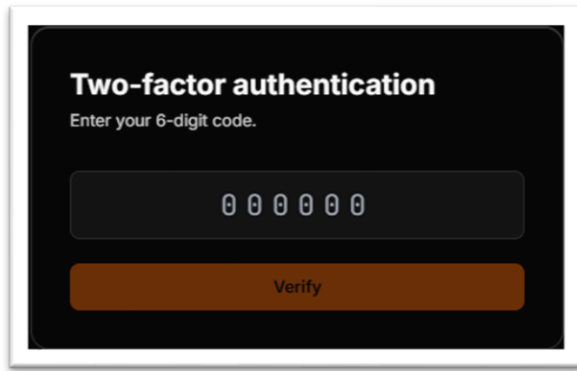
1. Enter your email and password on the sign-in page and select Sign in.

A screenshot of the EquityLens sign-in interface. The background is dark. At the top, it says "Welcome back" in white, followed by "Sign in to your EquityLens account". Below this are two input fields: "Email Address *" with the placeholder "your@email.com" and "Password *" with the placeholder "Enter your password". To the right of the password field is an eye icon. Below the password field is a link "Forgot password?". At the bottom is a large orange button labeled "Sign In". Below the button is a link "Don't have an account? Create one".

2. EquityLens shows a QR code. Open your authenticator app, choose “add account” (Or usually a + button), and point your phone’s camera at the QR code. If you can’t scan, select the option to reveal the setup key and type it into the app instead.
3. Your authenticator app now shows a 6-digit code for “Equity Lens” that changes every 30 seconds. Enter the current code on screen to prove the pairing worked.

Every subsequent sign-in:

1. Enter your email and password.
2. When prompted, open your authenticator app and enter the current 6-digit Equity Lens code.

A screenshot of a two-factor authentication interface. It features a dark background with white text. At the top, the title "Two-factor authentication" is displayed in a bold font, followed by the instruction "Enter your 6-digit code." Below this is a light gray input field containing six white circles, each representing a digit. At the bottom of the form is a prominent orange button labeled "Verify".

Two-factor authentication

Enter your 6-digit code.

0 0 0 0 0 0

Verify

If you type the code incorrectly or the code rolls over while you type the form tells you and lets you try again with the new code.

To sign out, use the sign-out control at the top-right of any page.

Finding Your Way Around

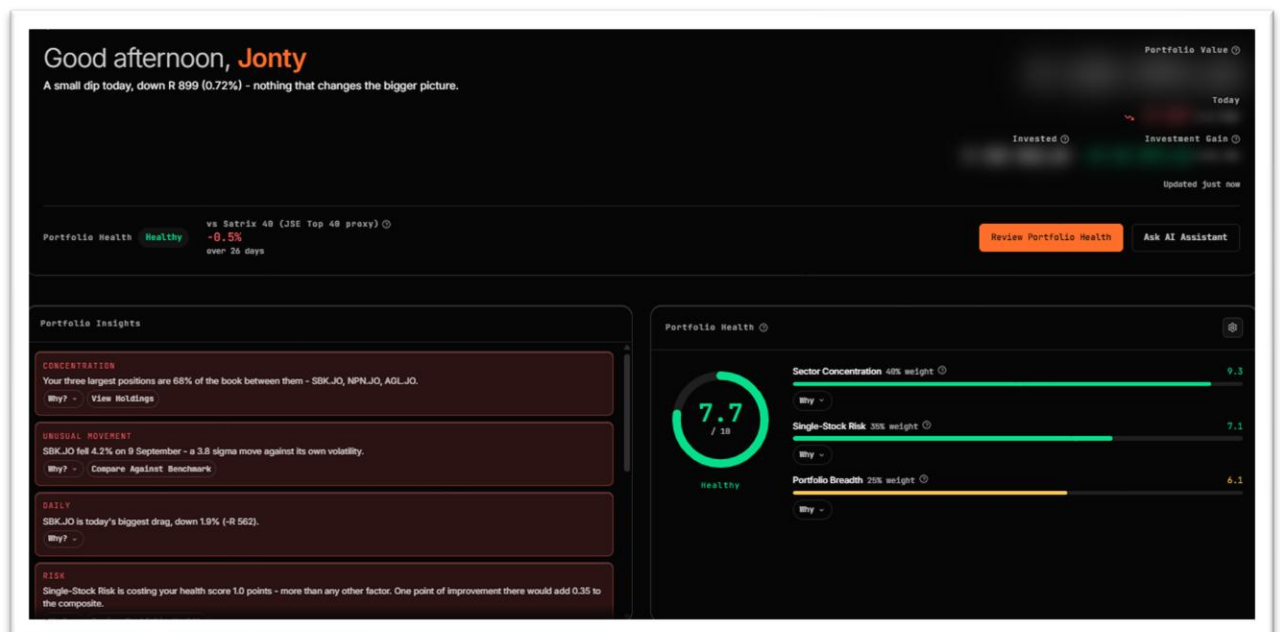
Once signed in, every page shares the same layout.

The sidebar (left edge. On a phone it is opened with the menu button) is how you move between the five areas of EquityLens:

- Dashboard - your portfolio (Section 7).
- Portfolio - your holdings in detail and where you import data (Section 8).
- Analytics - the financial indicators (Section 9).
- News - news stories connected to the stocks you own (Section 10).
- AI Assistant - ask questions about your portfolio (Section 11).
- Explore - JSE shares similar to the ones you hold (Section 12).
- Settings - theme and account management (Section 14).
- Help – the Help Center (Section 15).

The Topbar:

- Menu button - opens and closes the sidebar.
- Blur values - hides value amounts on screen in case of screen sharing.
- Sign out.



Dashboard

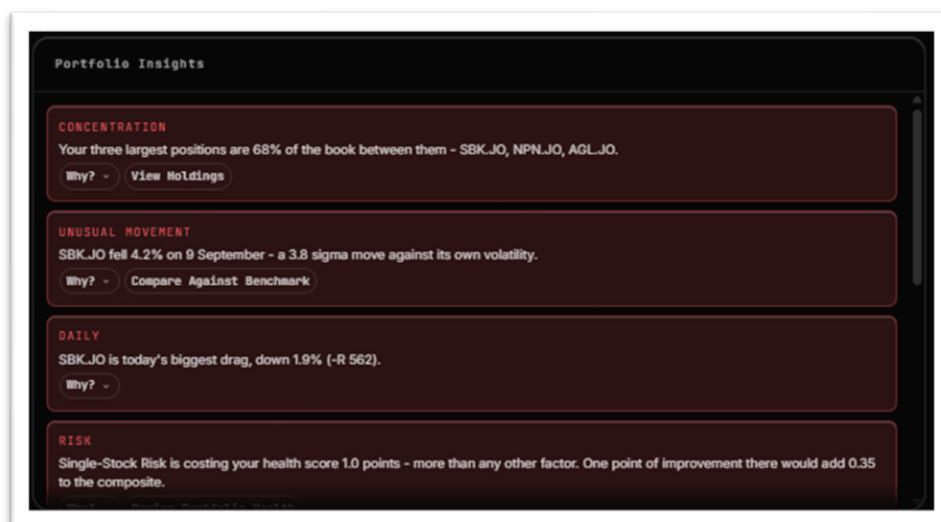
The Dashboard answers four questions in one screen: What am I worth? Am I healthy? How am I doing against the market? What needs my attention?

If your portfolio is empty, the Dashboard says so and points you to the Portfolio page to import your data (Section 8) and everything below comes alive once you have.

Overview. A greeting with your portfolio in rands, the amount you have invested and your investment gain. It also shows how much that moved today. It shows your portfolio health and buttons to review your portfolio health and asking the ai assistant.



Portfolio Insights. These are the things worth looking into. Each links you to the part of the dashboard that it concerns and can hand it to the AI assistant to explain.

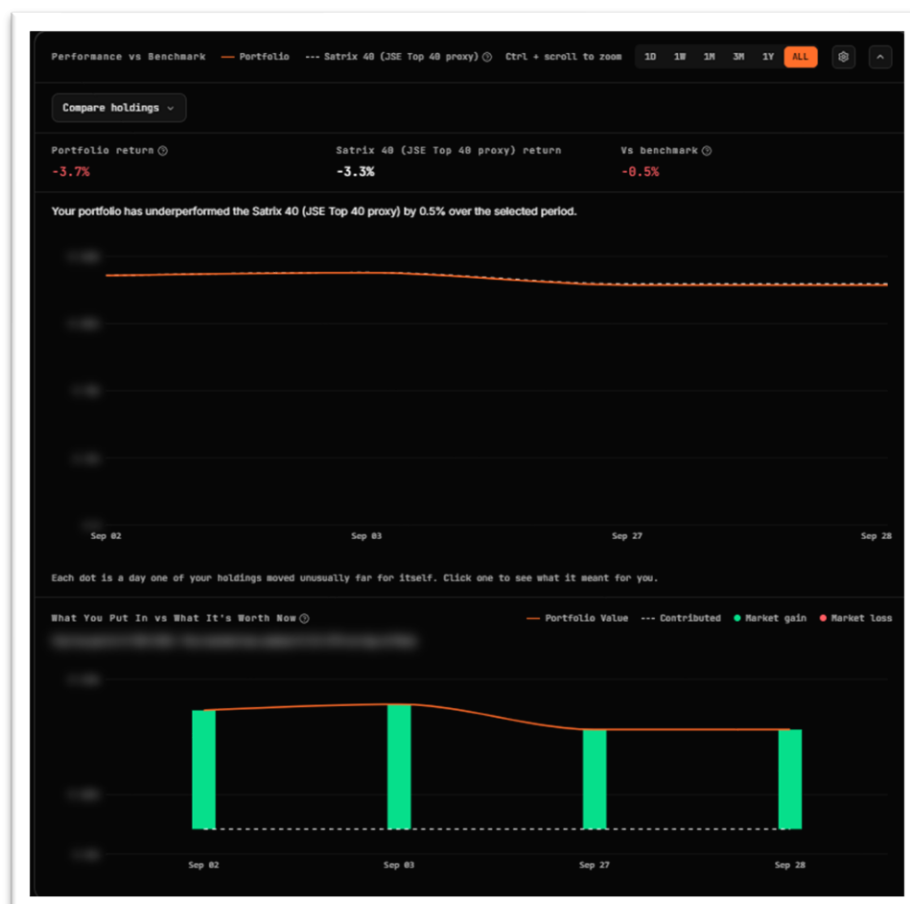


Portfolio health. A score out of 10 built from four ingredients you can see individually: Diversification of your holdings, how exposed you are to sector, your single-stock

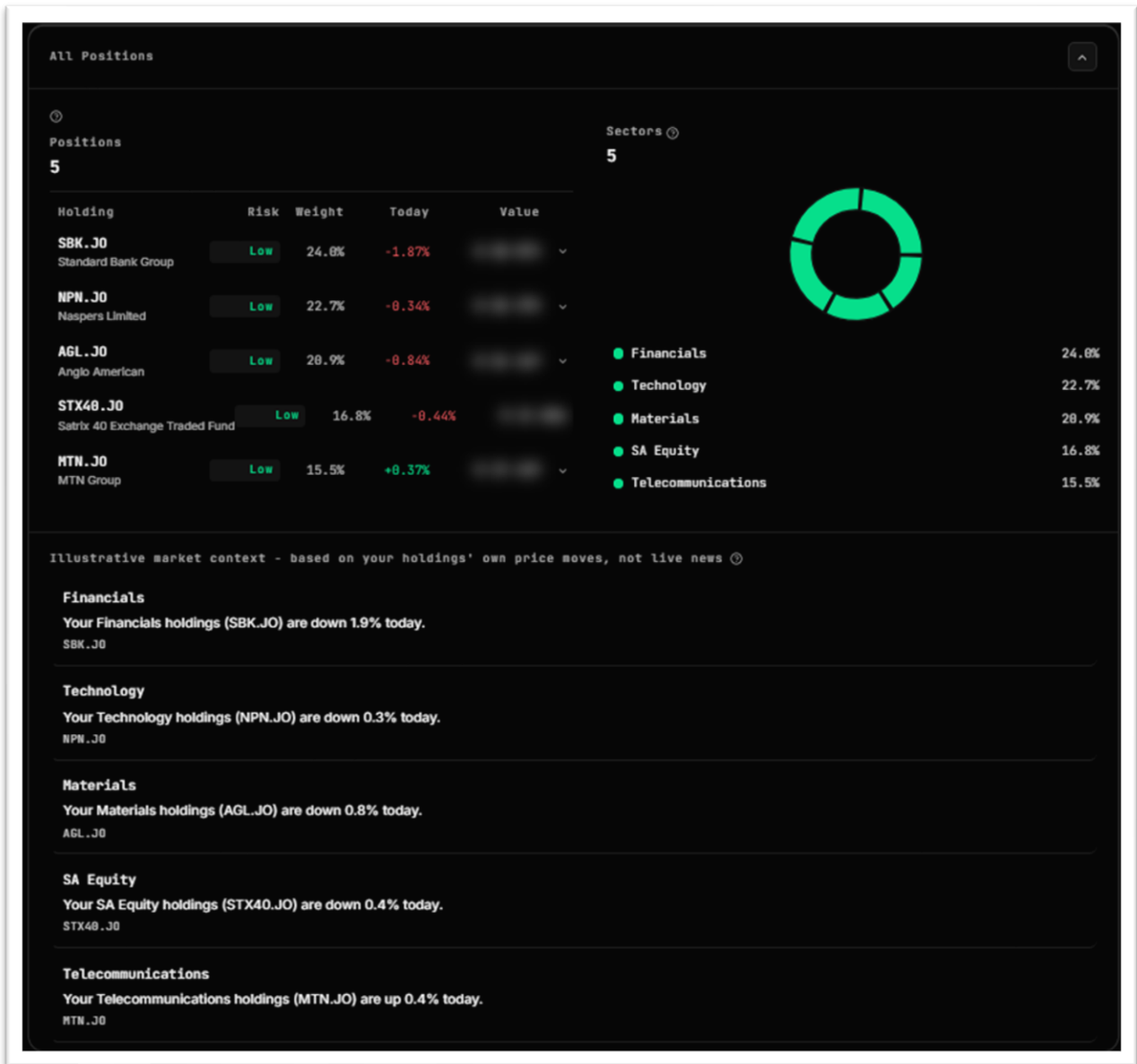
concentration risk, how many sectors your portfolio is stretched across and your performance against the benchmark.



Performance vs Benchmark. A chart of your portfolio against the JSE All Share Index (ALSI). Both lines start level at the beginning of the selected range, so you compare the shape of the move, not the size. For the period before you uploaded, EquityLens rebuilds your history from the transactions on your statement. The panel lists unusual rises and falls and expand one to see what drove it or ask the AI assistant about it.



All positions. Every holding with its weight, value, today's move, return total and a risk flag as well as a sector breakdown.



Concentration & Rebalancing. This is simulations at the sector level. You can see what adding to a sector will do for you portfolio health score.

Concentration & Rebalancing

Add to a sector

What a 5% new position in one sector would do to your Portfolio Health.

Telecommunications

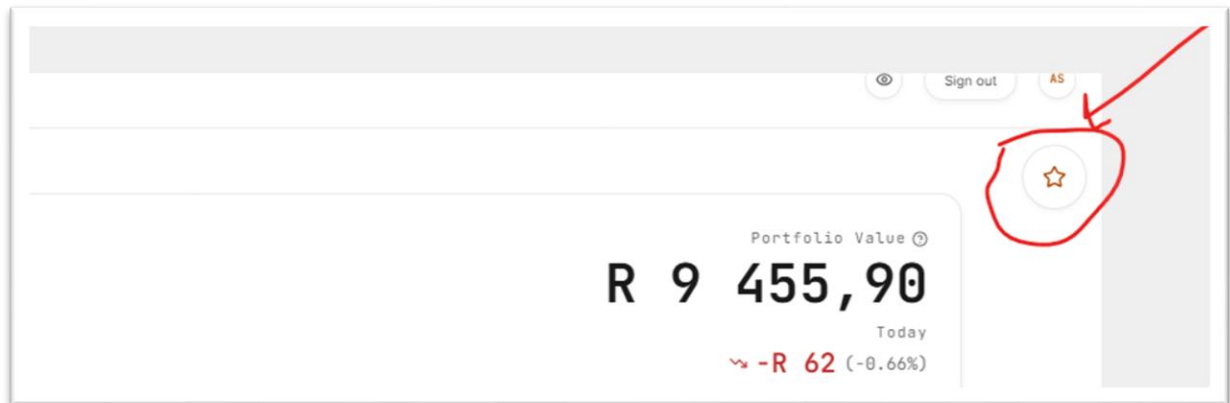
Simulate investing in Telecommunications

Rebalance your most concentrated sector

What moving the excess out of Financials and into Telecommunications would do.

Simulate shifting Financials into Telecommunications

Your Watchlist

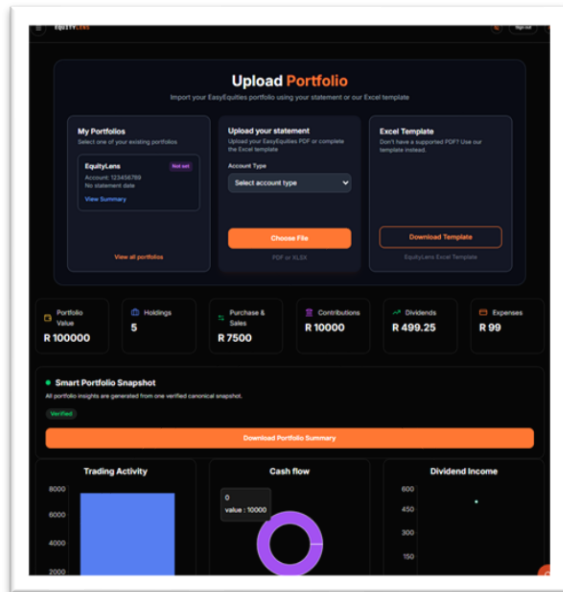


The watchlist is for stocks you want to keep an eye on.

- To add a stock: search by ticker or company name, pick the right match from the results, and it joins your list showing its current price and today's percentage change.
- To remove a stock: select the remove action on its entry.

If a watched stock's price can't be fetched now, that entry says so while the rest of the list stays current.

Portfolio



Open Portfolio in the sidebar. You can import as many portfolios as needed.

Importing a statement:

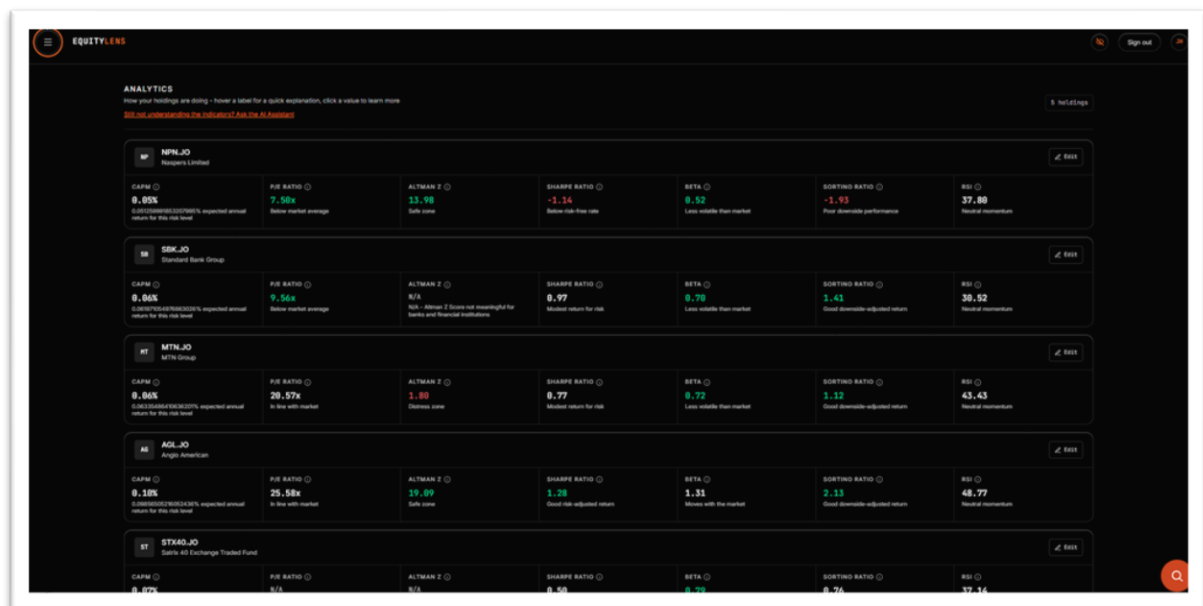
1. Choose the account type before choosing a file: ZAR, TFSA or USD.
2. Select your file by clicking the Choose File button. This can either be an EasyEquities PDF statement or download the excel template and fill that out.
3. If you choose to select a PDF then you must enter the password for that statement which would be the id number of that account.
4. EquityLens reads this file in your browser and then would appear under the My Portfolios section on the left.

Viewing a portfolio

Under the section My Portfolios, you can click view summary of the portfolios you have uploaded to view all different figures and charts.

Select Download Portfolio Summary to get a summary as a PDF report.

Analytics



Open Analytics in the sidebar. This page calculates financial indicators from your actual holdings and live market data which professionals use, each explained in plain language.

The seven indicators:

- Sharpe Ratio - are your returns worth the risk you're taking?
- Sortino Ratio - the same question but only counting downside risk (the kind that hurts).
- P/E Ratio - how expensive each share is relative to the company's earnings.
- CAPM Expected Return - what return the market's logic says a holding should deliver for its risk.
- Altman Z-Score - a company financial-health warning system: safe, grey zone, or distress.
- Beta - how much a holding swings compared to the JSE.
- RSI - whether a stock has recently been bought (or sold) unusually hard: overbought, neutral, or oversold.

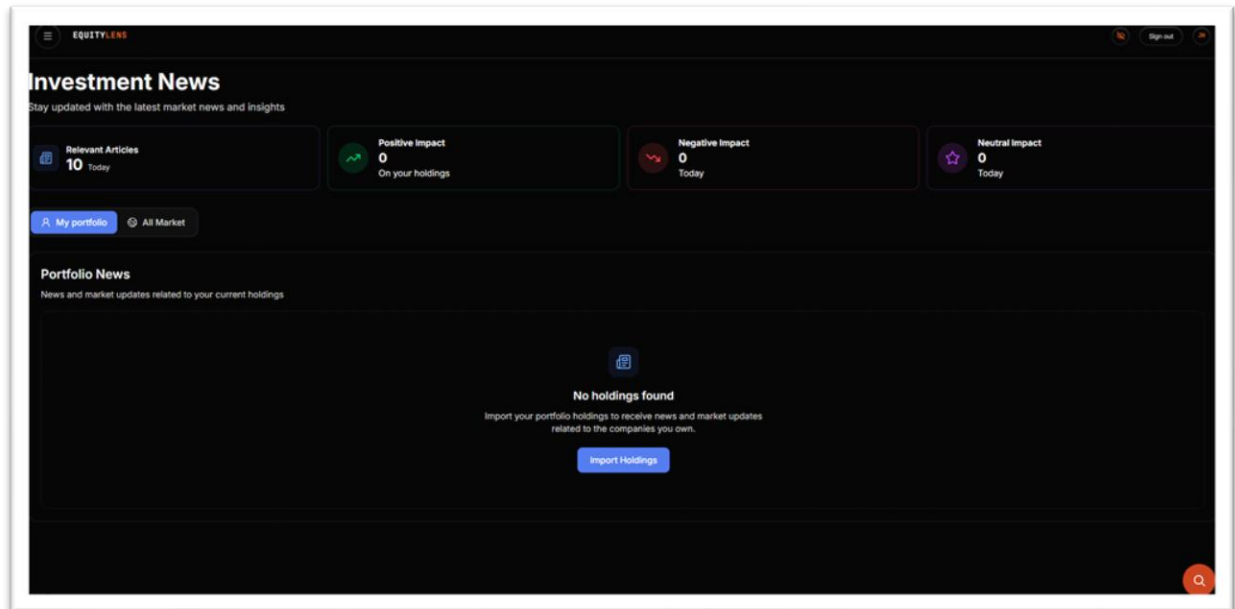
Hover over or click on any indicator's name and an explanation appears telling you what it means and how to read your number. Values are colour coded, positive, neutral, negative with using each indicator's standard thresholds.

If there isn't enough data to calculate an indicator for a holding its card says N/A with a short reason instead of showing a made-up number; and if a calculation fails, only that card shows an error.

Choosing which indicators you see. Select Customize Indicators on a holding to choose which of the seven indicators are shown for it. If you use the same selection

often, select Save Current Selection as Preset and give it a name. Your presets are listed under Your Presets so you can apply them again with one click.

News



Open News in the sidebar. The page has two tabs.

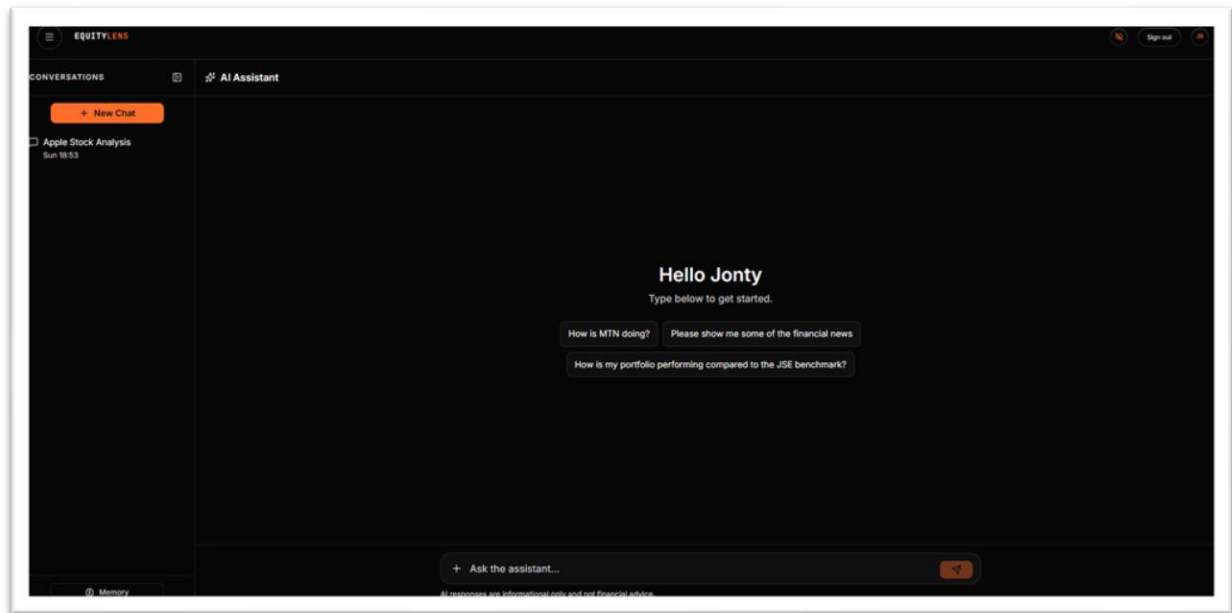
Portfolio News shows news and market updates related to the shares you hold. Filter it by one of your tickers to see only that company's stories. A summary shows how many of the stories are positive, negative or neutral.

All Market shows the latest financial and market stories, which you can narrow by category: Business, Top, Technology, Politics or Crime.

Each article shows its headline, a short description, its source and date, the related tickers, and a sentiment badge (positive, negative or neutral) describing the tone of the coverage. Selecting an article opens the full story at the original publication.

If the news service is temporarily unavailable, you will see a clear message instead of the list.

Using the AI Assistant



Open AI Assistant in the sidebar or use the floating assistant button available on every other page. This is where you ask questions about your portfolio.

Asking a question. Type into the box at the bottom ("Ask the assistant...") and press Enter. Questions can be up to 500 characters. The answer appears word by word as it is written. If you are unsure where to start, select one of the suggested prompts shown when a chat is empty.

What it can look up. Besides your portfolio, the assistant can fetch a share's latest price, recent news about a company, a share's indicators (the same ones as the Analytics page), the fees, dividends, trading and cash flow from your statements, and projections of how your portfolio might grow toward a goal. Prices are end-of-day, and projections are estimates with the assumptions stated.

Choosing a portfolio. If you have more than one portfolio you can use the portfolio selector beside the message box to choose which one the conversation is about or leave it on All portfolios.

Your conversations are saved. The panel beside the chat lists every conversation, newest first, each with an automatically generated title. Select one to pick up where you left off or select + New Chat to start fresh. To rename or delete a conversation you should hover over its name and select the pen or the bin icon.

Working with answers. Under each answer you can Copy it. Under the latest answer you can regenerate it to get a new version. If an answer fails, select retry.

Memory. The assistant remembers lasting facts you tell it across all your conversations. When it saves one, you will see "Added to memory". Select Memory to see everything it remembers and delete anything you don't want kept.

Limits. To keep the service fair, you can send up to 5 messages a minute and 50 a day. If you hit the limit, the message box shows how many seconds are left before you can send again.

The assistant will not tell you to buy or sell anything, and it only helps with EquityLens and investing questions.

If the assistant can't answer you will see a message and can simply try again

Exploring Similar Shares



Open Explore in the sidebar. Explore finds JSE listed shares that are similar to the ones you already have. EquityLens suggests up to nine shares with each suggestion shows its ticker and why it was picked: which of your holdings it is similar to, how close the match is as a percentage, and what it has in common. The Visual map plots your holdings alongside the suggestions so you can see how they relate to each other.

Things to know:

- Explore only covers JSE listed shares. Any holdings that aren't on the JSE are listed as "Not included".
- If your portfolio contains no JSE listed shares yet, Explore will say so instead of showing suggestions.
- Suggestions are for research and are not recommendations to buy. EquityLens never gives financial advice.

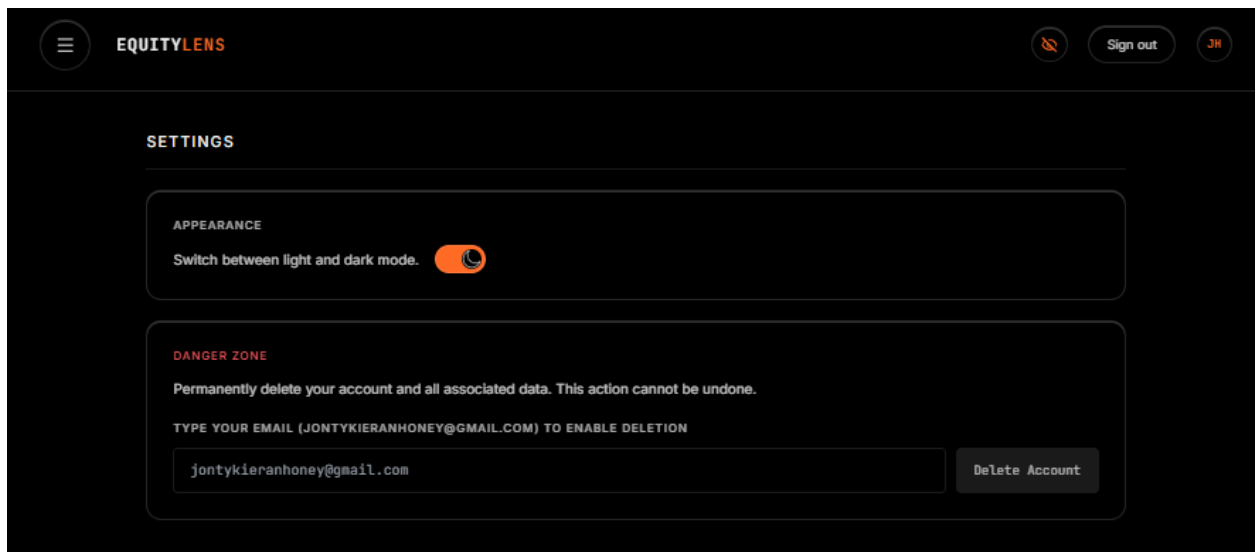
Settings

Open Settings in the sidebar.

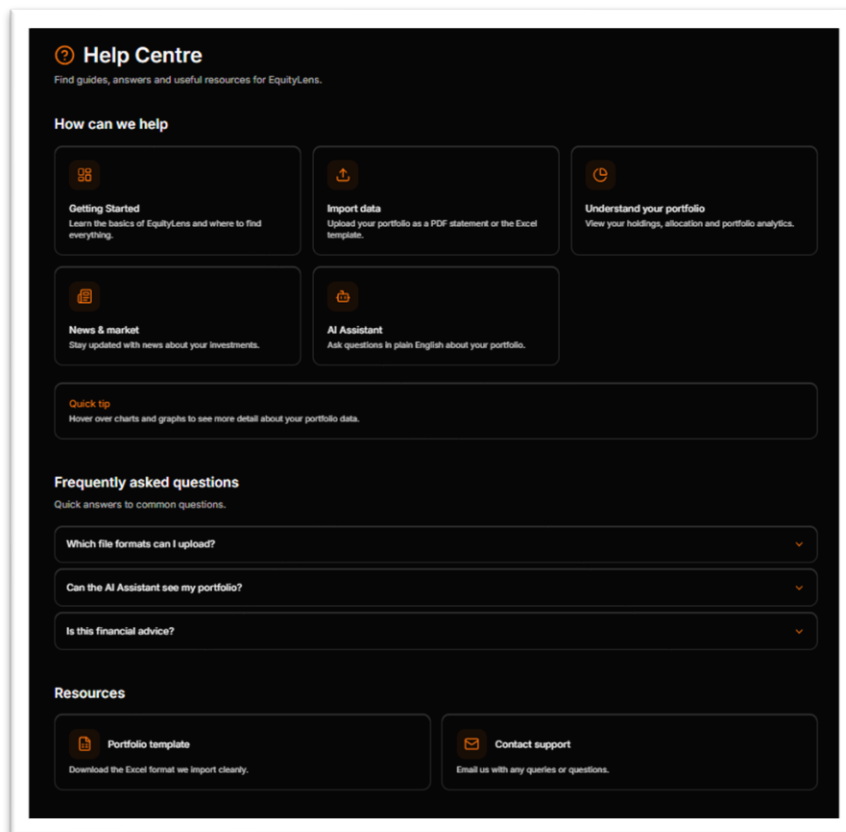
Theme. Switch between light and dark mode.

Delete Account. Permanently deletes your account and everything in it.

To confirm, type your email address to enable the button, then confirm once more.
This cannot be undone.



Getting Help



- The Help Centre - reached from the landing page navigation and from within the application's sidebar, located at the bottom, it contains tips on how to work and navigate this application
- Contact the team: thebigfivetb5@gmail.com .
- It also contains Troubleshooting and Frequently asked questions.
 - o These include signing up and signing in.
 - o Importing
 - o The AI assistant
 - o Analytics and explore
 - o Frequently asked questions

Glossary

Term	Meaning
ALSI (All Share Index)	The index of all main JSE shares which is EquityLens's benchmark.
Altman Z-Score	A measure of a company's financial health: safe, grey zone or distress.
Benchmark	What your performance is compared against.
Beta	How much a share swings compared to the market as a whole. Above 1 swings more or below 1 swings less.
CAPM expected return	The return a share should deliver for the risk it carries.
Cost basis (Invested)	What you paid for the holdings you currently own.
Diversification	Spreading your money across many companies and sectors so one bad result can't sink the whole portfolio.
Dividend	A share of a company's profit paid out to shareholders.
Holding / position	A share you own and how many of it.
JSE	The Johannesburg Stock Exchange.
P/E ratio	A share's price divided by the company's earnings per share; how expensive the share is relative to its profits.
Portfolio breadth	How many holdings you effectively have once their sizes are taken into account.
Rebalancing	Moving money between holdings or sectors to change how your portfolio is spread.
RSI (Relative Strength Index)	Whether a share has recently been bought or sold unusually hard: overbought, neutral or oversold.
Sector	The part of the economy a company operates in, such as Financial Services or Healthcare.
Sentiment	Whether news coverage of a company reads as positive, negative or neutral.
Sharpe ratio	Return earned per unit of risk taken.
Sortino ratio	Like the Sharpe ratio, but only counts downside risk.
TFSA	Tax-Free Savings Account.

Ticker	The short code a share trades under, e.g. NPN.JO for Naspers on the JSE.
Time-weighted return	Growth measured so that deposits and withdrawals don't count as gains or losses.
Two-step verification	Signing in with your password plus a code from your authenticator app.
Unrealised gain	Profit on shares you still hold, which you haven't locked in by selling.
Withholding tax	Tax deducted from a dividend before it is paid to you.